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INVITED TALK / ABSTRACT

Inverse mean curvature flow and applications

Alessandra Pluda

Università di Pisa

Abstract

A classical solution to the inverse mean curvature flow is a time-dependent family of smooth immersions whose normal velocity, at each point and time, equals the inverse of the mean curvature. Over the years, the inverse mean curvature flow has proven to be an extremely useful tool, ranging from the proof of the Riemannian Penrose inequality to the characterization of Ricci-pinched 3-dimensional Riemannian manifolds. In this talk, I will focus on the inverse mean curvature flow as an instrument to prove geometric inequalities, from the isoperimetric inequality to a family of Willmore/Minkowski-type inequalities.

Keywords: inverse mean curvature flow; mean curvature; Riemannian Penrose inequality; Ricci-pinched Riemannian manifolds; geometric inequalities; isoperimetric inequality; Willmore/Minkowski-type inequalities